INCH-POUND

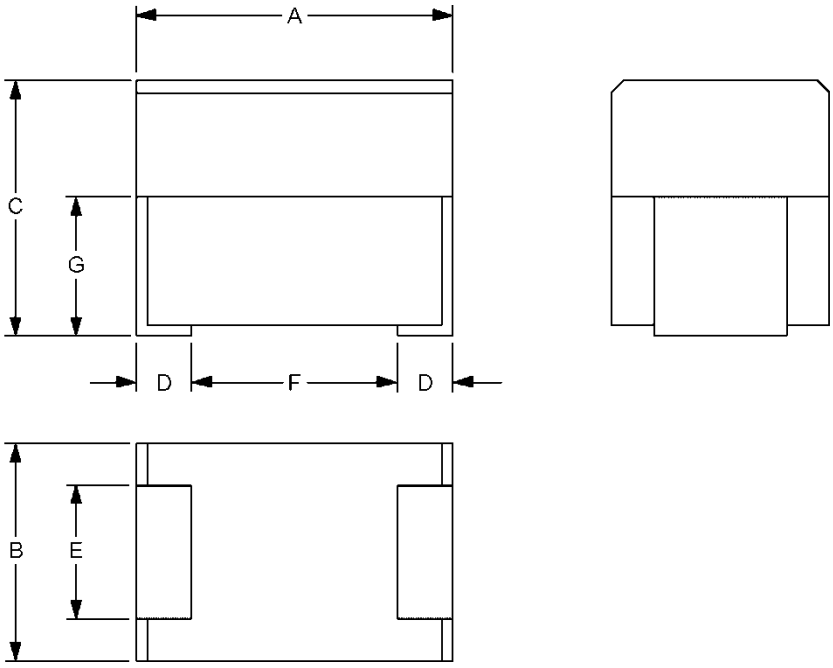
MIL-PRF-27/368B
12 April 2013
SUPERSEDING
MIL-PRF-27/368
14 July 2006

PERFORMANCE SPECIFICATION SHEET

INDUCTOR, POWER, SURFACE MOUNT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-27.




| | Inches | mm |
|---|---|---|
| A | .166 to .190 | 4.22 to 4.83 |
| B | .118 to .134 | 3.00 to 3.40 |
| C | .118 to .138 | 3.00 to 3.51 |
| D | .015 Min. | 0.38 Min. |
| E | .054 to .078 | 1.37 to 1.98 |
| F | .118 (Ref. Only) | 3.00 (Ref. Only) |
| G | .066 (Ref. Only) | 1.68 (Ref only) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configuration.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:24Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics (initial).

| Dash number | Inductance (μH) ±10% 1/ | DC resistance ohms (max) | Current rating (mA DC) (max) | Incremental current (mA DC) 2/ |
|---|---|---|---|---|
| -01 | 1.0 | 0.113 | 1050 | 2400 |
| -02 | 1.2 | 0.199 | 1000 | 2158 |
| -03 | 1.5 | 0.222 | 950 | 1980 |
| -04 | 1.8 | 0.240 | 900 | 1828 |
| -05 | 2.2 | 0.268 | 850 | 1697 |
| -06 | 2.7 | 0.288 | 800 | 1513 |
| -07 | 3.3 | 0.323 | 750 | 1397 |
| -08 | 3.9 | 0.347 | 700 | 1250 |
| -09 | 4.7 | 0.401 | 650 | 1131 |
| -10 | 5.6 | 0.437 | 650 | 1060 |
| -11 | 6.8 | 0.472 | 600 | 990 |
| -12 | 8.2 | 0.548 | 600 | 871 |
| -13 | 10 | 0.608 | 550 | 772 |
| -14 | 12 | 0.670 | 550 | 752 |
| -15 | 15 | 0.780 | 450 | 693 |
| -16 | 18 | 0.875 | 400 | 614 |
| -17 | 22 | 1.200 | 370 | 515 |
| -18 | 27 | 1.404 | 330 | 465 |
| -19 | 33 | 1.578 | 300 | 425 |
| -20 | 39 | 1.848 | 280 | 376 |
| -21 | 47 | 2.064 | 260 | 366 |
| -22 | 56 | 2.268 | 240 | 336 |
| -23 | 68 | 3.408 | 220 | 297 |
| -24 | 82 | 3.648 | 200 | 267 |
| -25 | 100 | 4.320 | 180 | 238 |
| -26 | 120 | 6.000 | 160 | 208 |
| -27 | 150 | 6.432 | 140 | 183 |
| -28 | 180 | 8.400 | 120 | 158 |
| -29 | 220 | 9.680 | 120 | 143 |
| -30 | 270 | 12.720 | 100 | 129 |
| -31 | 330 | 15.240 | 90 | 118 |

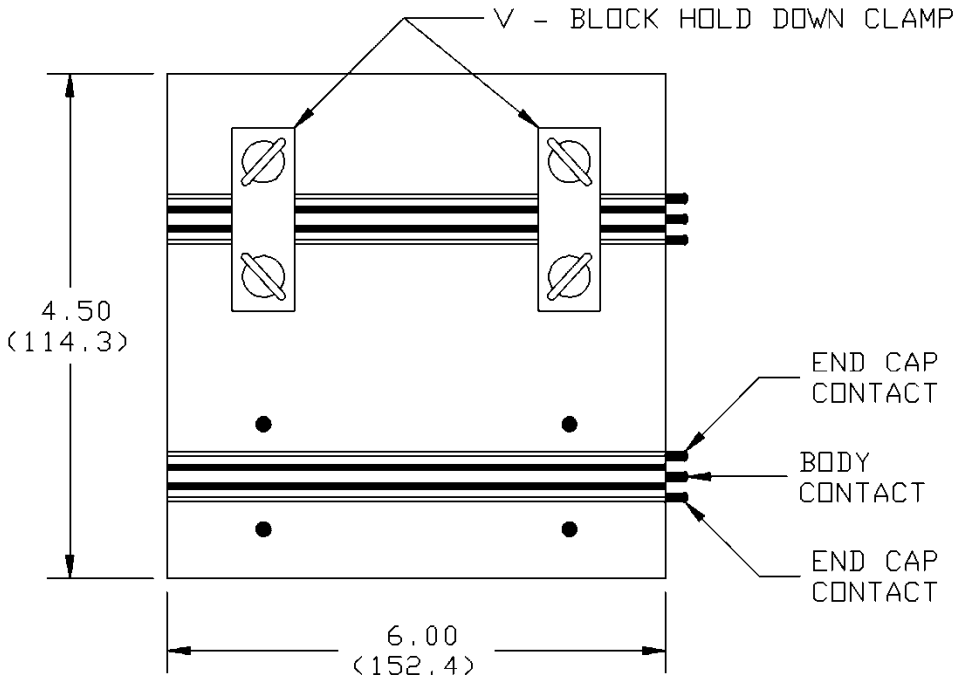
1/ Measured at 1 V with no dc current at 1 KHz.
2/ The dc current at which the inductance will decrease by a maximum of 5 percent from the inductance at zero dc current.

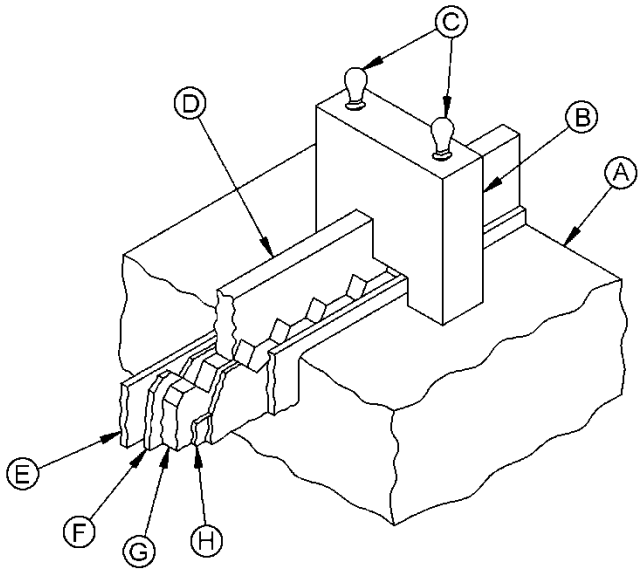
TABLE II. Electrical characteristics (final).

| Inspection group | Allowable percent variation from the initial measurement | |
|---|---|---|
| Qualification inspection: | Inductance | DC resistance |
| Group II | ±5 | ±3 (+.001 ohm) |
| Group III | ±5 | ±2 (+.001 ohm) |
| Group IV | ±5 | ----- |



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:24Z
Check the source to verify that this is the current version before use.




ENLARGED CUT-AWAY OF CLAMP

| Legend | |
|---|---|
| LTR | Description |
| A | G-10 base |
| B | Upper V-block hold-down clamp |
| C | Clamp screws |
| D | Upper V-block (body contact) |
| E | End cap contact retaining wall |
| F | End cap contact V-block (in contact with E) |
| G | Lower V-block (body contact) |
| H | G-10 insulators (separating G from F/E) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 2. Test fixture for insulation resistance, dielectric withstanding voltage, and barometric pressure.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:24Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 1.5 grams maximum.

Case: Grade 5, encapsulated.

Operating temperature range: -55°C to +130°C, Class S.

Temperature rise (at +90°C): +40°C, method of mounting shall be on test substrates that provide test pieces one inch minimum separation.

Altitude: 70,000 feet.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202. Depth of immersion shall be the entire mounting surface for 4 to 5 seconds.

Bond strength: Method 2011, test condition F, MIL-STD-883.

Force: 2 pounds.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 200 V rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202, (70,000 feet), test voltage 80 V rms.

Insulation resistance: Method 302, MIL-STD-202, 1000 megohms minimum at 100 V dc.

Electrical characteristics (initial): See table I.

Vibration, high frequency: Method 204, test condition D, MIL-STD-202.

Moisture resistance: Method 106, MIL-STD-202, method of mounting shall be on test substrates that provide test pieces one inch minimum separation.

Test fixture: Test shall be performed using test fixture shown on figure 2, or equivalent.

Quality Assurance Provisions:

Extent of qualification: Qualification testing and approval to M27/367-61 shall be sufficient to grant qualification to: M27/367-01 through M27/367-61 inclusive and M27/368-01 through M27/368-31 inclusive and M27/370-01 through M27/370-52 inclusive, all parts.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/368-(dash number from table I).

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202
MIL-STD-883



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:24Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2013-023)

Review activity:
Army – MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:24Z
Check the source to verify that this is the current version before use.